PHARMACOLOGY

GIT Infection and Antiviral Drugs

SGL 3
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GIT BLOCK

Pharmacology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. What is the single greatest danger of any GI infection, and what is the mainstay treatment for it?
2. Which antibiotic is repeatedly favored for GI bacterial infections in children, pregnancy, and breastfeeding?
3. What is the key pharmacokinetic feature that makes sofosbuvir activate specifically in the liver?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 GI infections — overview & oral rehydration therapy

The single greatest danger in any GI infection is **dehydration** — oral rehydration therapy (ORT) is the priority intervention. Enteric **viral** infections (rotavirus — the most common cause, presenting with moderate-severe vomiting then watery diarrhoea and fever; also norovirus, astrovirus, enteric adenovirus) have **no specific antiviral drug**: treatment is conservative (antipyretics, antiemetics, ORS).

Component (per litre)	Original ORS	Low-osmolarity ORS*
Glucose, anhydrous	20.0 g	13.5 g
Sodium chloride	3.5 g	2.6 g
Trisodium citrate	2.9 g	2.9 g
Potassium chloride	1.5 g	1.5 g

*Low-osmolarity ORS has been the WHO/UNICEF-recommended formula since 2002 — it reduces stool output, vomiting, and need for IV therapy versus the original formula. **Source note:** the deck's earlier slide on "sodium bicarbonate based ORS" gives a different, older composition (NaCl 3.5g / glucose 20.0g / KCl 2.5g / sodium bicarbonate 2.5g per litre) that does not match this WHO/UNICEF low-osmolarity table — this summary uses the WHO/UNICEF figures as the authoritative current standard.

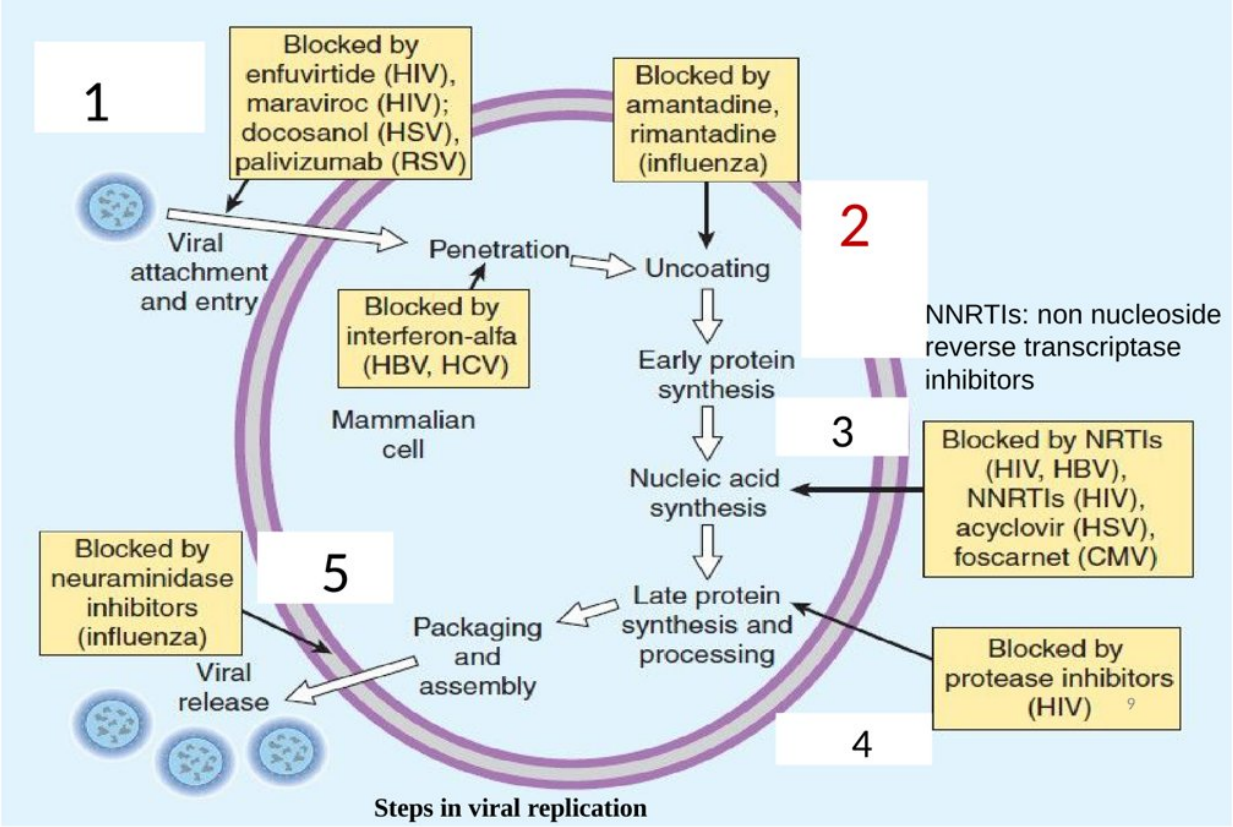
2 Bacterial GI infections — antibiotic regimens

Non-typhoid Salmonella	Ciprofloxacin 500mg/12h PO 5-7d (contraindicated <14y, pregnancy, breastfeeding) **or** ceftriaxone 1g IV/24h (safe in these groups)
Traveler's diarrhoea (ETEC/EPEC)	Azithromycin 500mg/24h PO 1-3d (safe <14y, pregnancy, breastfeeding); ciprofloxacin and rifaximin 200mg/8h PO 3d are alternatives
Shigella spp	Ciprofloxacin first-line (WHO); ceftriaxone 1g IV/24h or azithromycin 500mg/24h 1-3d second-line if first-line fails
Campylobacter jejuni	Azithromycin 500mg/24h PO 3 days (WHO-recommended)
Vibrio cholerae	Doxycycline 300mg single dose first-line; azithromycin 500mg/24h 3d as the alternative safe in pregnancy and children
C. difficile / pseudomembranous colitis	Stop the causative antibiotic. Oral metronidazole 250mg/6h first-choice; oral vancomycin 125mg/6h if metronidazole fails
Cholecystitis / liver abscess	Piperacillin/tazobactam, meropenem, imipenem/cilastatin, tigecycline, cefazolin/cefepime, or ciprofloxacin/levofloxacin

3 Viral hepatitis — overview & chronic HBV/HCV treatment goal

Hepatitis A and E are typically acute/self-limited (faecal-oral spread); **hepatitis B** (DNA virus) and **hepatitis C** (RNA virus) are the leading causes of **chronic hepatitis, cirrhosis, and hepatocellular carcinoma (HCC)**. Goal of chronic HBV/HCV therapy: **improve survival and quality of life** by preventing disease progression and HCC — not guaranteed complete eradication in every patient.

4 Antiviral drug classes — site of action



Steps in viral replication and where each antiviral class acts — interferon-alfa blocks HBV/HCV at penetration/uncoating; nucleoside/nucleotide analogues (NRTIs) and protease inhibitors act further downstream at nucleic-acid and late protein synthesis.

5 Key antiviral agents — pharmacology

Interferon-alpha / peg-IFN-alpha	Blocks viral penetration/uncoating. Pegylation ↑ size → delayed absorption, longer duration, ↓clearance → once-weekly dosing. Flulike symptoms/transient LFT rise/alopecia/rash/diarrhoea subside with continued use; dose-limiting: bone marrow suppression, severe fatigue/weight loss, autoimmune disorders. Caution in psychiatric disease, epilepsy, thyroid disease (risk of depression).
Lamivudine	Phosphorylated to active triphosphate; competitively inhibits HBV RNA-dependent DNA polymerase (reverse transcriptase) and HIV reverse transcriptase. Resistance can develop with prolonged use.
Tenofovir	Highly effective, low resistance vs adefovir; retains activity against lamivudine- and entecavir-resistant isolates.
Ribavirin	Used with peg-IFN for HCV. Oral bioavailability 64%, ↑ with high-fat meals, ↓ with antacids. Causes dose-dependent **haemolytic anaemia**; contraindicated in anaemia, end-stage renal failure, severe heart disease, pregnancy.
Sofosbuvir	Prodrug, triphosphorylated inside hepatocytes — activating enzymes present in liver cells, so activation occurs during first-pass metabolism right at its target organ. Can be used **without interferon** — valuable where IFN is contraindicated or was stopped for adverse effects.
Protease inhibitors (telaprevir, atazanavir)	Block viral protease. Telaprevir: take with food (↑ absorption); SEs — anaemia, rash, lipodystrophy, liver toxicity, diabetes; combined with peg-IFN + ribavirin to boost response.
Vosevi (sofosbuvir + velpatasvir + voxilaprevir)	Fixed-dose combo, approved July 2017, for HCV without cirrhosis or with mild (compensated) cirrhosis. Sofosbuvir + velpatasvir block two different viral RNA polymerase enzymes; voxilaprevir blocks the protease enzyme — 3 essential proteins blocked. **Must not** be co-administered with rosuvastatin, dabigatran, contraceptives, rifampicin, carbamazepine, phenobarbital, or phenytoin.

P Must-remember pearls

- There is no specific antiviral drug for enteric viral infections (rotavirus, norovirus, etc.) — management is purely conservative: antipyretics, antiemetics, and ORS.
- Azithromycin is the recurring 'safe in pregnancy/children/breastfeeding' choice across non-typhoid Salmonella, traveler's diarrhoea, and Campylobacter — while ciprofloxacin is repeatedly flagged as contraindicated in children <14y, pregnancy, and breastfeeding.
- Low-osmolarity ORS (WHO/UNICEF, since 2002) reduces stool output, vomiting, and need for IV therapy compared to the original high-osmolarity formula.
- C. difficile colitis: stop the causative antibiotic first; oral metronidazole is first-choice, oral vancomycin if metronidazole fails.
- Goal of chronic HBV/HCV therapy is NOT guaranteed eradication — it is to improve survival and quality of life by preventing progression to cirrhosis/HCC.
- Pegylation of interferon-alpha increases molecular size → delayed absorption, longer duration, decreased clearance → once-weekly dosing instead of 3x/week.
- Sofosbuvir is a prodrug triphosphorylated inside hepatocytes — activation happens during first-pass metabolism, right at its target organ.
- Ribavirin causes dose-dependent hemolytic anaemia and is contraindicated in anaemia, severe renal/cardiac disease, and pregnancy; its oral bioavailability rises with high-fat meals and falls with antacids.

Q Test yourself

1. A pregnant woman develops cholera. Which antibiotic is named as an appropriate alternative regimen for adults, including pregnant women and children?

A. Azithromycin 500 mg/24h orally for 3 days
B. Doxycycline 300 mg orally as a single dose
C. Ciprofloxacin 500 mg/12h orally for 3 days
D. Metronidazole 250 mg/6h orally for 7 days
E. Ceftriaxone 1g IV/24h for 5 days

2. A patient develops pseudomembranous colitis after a course of clindamycin. What is the first-choice treatment?

A. Oral vancomycin 125 mg/6h
B. IV ceftriaxone 1g/24h
C. Oral metronidazole 250 mg/6h
D. Oral ciprofloxacin 500 mg/12h
E. IV meropenem

3. A patient with chronic HBV infection is started on pegylated interferon-alpha instead of standard interferon-alpha. What is the main pharmacokinetic advantage of pegylation?

A. It eliminates the need for parenteral administration entirely
B. It increases the molecule's size, delaying absorption and lengthening duration of action, allowing once-weekly dosing
C. It increases hepatotoxicity to enhance antiviral effect
D. It shortens the half-life for more frequent, titratable dosing
E. It converts the drug into an oral formulation

4. A patient taking lamivudine for chronic HBV infection develops lactic acidosis. What is lamivudine's mechanism of action?

A. It is a protease inhibitor that blocks viral polyprotein cleavage
B. It is a cytokine that triggers host immune defenses against the virus
C. It directly blocks viral attachment and entry into hepatocytes
D. It is phosphorylated to its triphosphate form and competitively inhibits HBV RNA-dependent DNA polymerase (reverse transcriptase)
E. It inhibits host DNA polymerase to indirectly suppress viral replication

5. A patient with chronic HCV is prescribed ribavirin in combination with pegylated interferon alfa-2b. What adverse effect must be monitored closely?

A. Irreversible hepatotoxicity requiring transplant
B. Severe nephrotoxicity requiring dialysis
C. Acute pancreatitis requiring hospitalization
D. Severe QT prolongation requiring cardiac monitoring
E. Dose-dependent hemolytic anemia

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — A — Azithromycin 500 mg/24h orally for 3 days is specifically named as an appropriate alternative regimen for adults, including pregnant women, in V. cholerae infection; doxycycline and ciprofloxacin are the other options but are not highlighted as pregnancy-safe.

2 — C — Oral metronidazole 250 mg/6h is the first-choice treatment for C. difficile colitis; oral vancomycin is used if metronidazole fails. Stopping the causative antibiotic is also essential.

3 — B — Pegylation (attaching polyethylene glycol) increases the interferon molecule's size, delaying absorption from the injection site, lengthening duration of action, and decreasing clearance — allowing once-weekly rather than three-times-weekly dosing.

4 — D — Lamivudine is phosphorylated by host cellular enzymes to its active triphosphate form, which competitively inhibits HBV RNA-dependent DNA polymerase (reverse transcriptase) and HIV reverse transcriptase.

5 — E — Ribavirin causes dose-dependent hemolytic anemia; it is contraindicated in anemia, end-stage renal failure, severe heart disease, and pregnancy.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam

Pharmacology GIT Block · MEDucation by Mattin Majid · Source: Dr. Jangi Shawkat Salai lecture slides, SGL 3 (full 42-slide deck); references WHO Drug Information Vol. 16 No. 2 (2002).
Study alongside the lecture, not instead of it.